STAT 6410
Homework 2

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1. Appendix A, Problem 29 Let $\mathbb{S} = C[0,1]$ and $d_p(f,g) \equiv (\int_0^1 |f(t) - g(t)|^p dt)^{1/p}$ for $1 \leq p < \infty$ and $d_\infty(f,g) = \sup\{|f(t) - g(t)| : t \in [0,1]\}$.

   (a) Let $f(x) \equiv 1$. Let $f_n(t) \equiv 1$ for $0 \leq t \leq 1 - \frac{1}{n}$, and $f_n(t) = n(1-t)$ for $1 - \frac{1}{n} \leq t \leq 1$. Show that $d_p(f_n, f) \to 0$ for $1 \leq p < \infty$ but $d_\infty(f_n, f) \nrightarrow 0$

   (b) Fix $f \in C[0,1]$. Let $g_n(t) = f(t)$, $0 \leq t \leq 1 - \frac{1}{n}$, and $g_n(t) = f(1-\frac{1}{n}) + (f(1) - f(1-\frac{1}{n}))n(t + \frac{1}{n} - 1)$, $1 - \frac{1}{n} \leq t \leq 1$.

   Show that $d_p(g_n, f) \to 0$ for all $1 \leq p \leq \infty$.

   You may use this inequality: $|a+b|^p \leq 2^p(|a|^p + |b|^p)$

Ans: (a)
For $1 \leq p < \infty$:

$$d_p(f_n, f) = \left(\int_0^{1-\frac{1}{n}} |1-1|^p dt + \int_{1-\frac{1}{n}}^1 |1 - n(1-t)|^p dt\right)^{1/p}$$

$$= \left(\int_{1-\frac{1}{n}}^1 |1 - n + nt|^p dt\right)^{1/p}$$

$$(d_p(f_n, f))^p = \int_{1-\frac{1}{n}}^1 |1 - n + nt|^p dt$$

$$= \int_{1-\frac{1}{n}}^1 (1 - n + nt)^p dt \qquad 1 - n + nt > 0$$

Define: $u = 1 - n + nt \quad du = n\, dt$

$$= \frac{1}{n}\int_0^1 u^p du = \frac{u^{p+1}}{n(p+1)}\Big|_0^1 = \frac{1}{n(p+1)}$$

$$\lim_{n\to\infty} \frac{1}{n(p+1)} = 0$$

Since $\lim_{n\to\infty} d_p(f_n, f) = 0$, $\left(\lim_{n\to\infty} d_p(f_n, f)\right)^p = 0$

For $p = \infty$

$$\begin{aligned}
&d_\infty(f_n, f) = \sup\{|f_n(t) - f(t)| : t \in [0,1]\} \\
&\textbf{Case 1: } 0 \le t \le 1 - \tfrac{1}{n} \\
&|f_n(t) - f(t)| = |1 - 1| = 0 \\
&\textbf{Case 2: } 1 - \tfrac{1}{n} \le t \le 1 \\
&|n(1-t) - 1| = |n - nt - 1| \\
&\lim_{n\to\infty} \sup\{|n - nt - 1\} = 1
\end{aligned}$$

Since $\lim\limits_{n\to\infty} d_\infty(f_n, f) = 1$, $d_\infty(f_n, f) \nrightarrow 0$.

(b)

$$d_p(f_n, f) = \left(\int_0^{1-\frac{1}{n}} |1-1|^p dt + \int_{1-\frac{t}{n}}^{1} 1 + (1-1)n(t + \frac{1}{n} - 1)dt \right)$$

$$= (0 + 1|_{1-\frac{1}{n}}^{1} = 1 - 1 = 0$$

$$\lim_{n\to\infty} d_p(f_n, f) = \lim_{n\to\infty} 0 = 0$$

2. Appendix A, Problem 34

Show that unions of open sets are open and intersection of any two open sets is open. Give an example to show that the intersection of an infinite number of open sets need not be open.

Ans: Let $(\mathbb{S}, d)$ be a metric space.

WTS: The union of open sets are open.

Proof. Let $I \in \mathbb{R}$ be an index set, and $\{A_j\}_{j\in I}$ be a subset of open sets on $(\mathbb{S}, d)$.

Let $x \in \cup_{i\in I} A_i$ be given.

Thus, $x \in A_i$ for some $i \in I$. Since A_i is an open set, there exists $\epsilon > 0$ such that for all $y \in \mathbb{S}$ satisfying $d(x,y) < \epsilon$, $y \in A_i \subset \cup_{j\in I} A_j$.

Therefore, by definition, the unions of open sets are open. □

WTS: The intersection of any 2 open sets is open.

Proof. Let A_1 and A_2 be open sets on $(\mathbb{S}, d)$.
Thus,

- For any $y_1 \in A_1$, there exist $\epsilon_1 > 0$ such that for all $x_1 \in \mathbb{S}$ satisfying $d(x_1, y_1) < \epsilon_1$ belongs to A_1.
- For any $y_2 \in A_2$, there exist $\epsilon_2 > 0$ such that for all $x_2 \in \mathbb{S}$ satisfying $d(x_2, y_2) < \epsilon_2$ belongs to A_2.

Let $y \in A_1 \cap A_2$ be given. Thus, there exists $\epsilon_1 > 0$ such that all $x_1 \in \mathbb{S}$ satisfying $d(x_1, y) < \epsilon_1$ belong to A_1 and there exist $\epsilon_2 > 0$ such that all $x_2 \in \mathbb{S}$ satisfying $d(x_2, y_2) < \epsilon_2$ belong to A_2.

Thus, all $z \in \mathbb{S}$ satisfying $d(z, y) < \min(\epsilon_1, \epsilon_2)$ belong in $A_1 \cap A_2$.

Thus, the intersection of any 2 open sets is open. □

WTS: an example to show that the intersection of an infinite number of open sets need not be open

Let $A_n = (-\frac{1}{n}, \frac{1}{n})$, which is an open set on $(\mathbb{R}, |\cdot|)$.

$\cap_{n \in (0,\infty)} A_i = \{0\}$ which is not an open set. Thus, this is an example of an intersection of an infinite number of open sets that is not open.

3. Appendix A, Problem 36

   Let $f_n(x) = x^n$ and $g(x) \equiv 0$ on $\mathbb{R}$. Then $\{f_n\}_{n\geq 1}$ converges pointwise to g on $(-1, 1)$, uniformly on $[a, b]$ for $-1 < a < b < 1$, but not uniformly on $(0, 1)$.

Ans:

Proof. **Part 1:** WTS: $\{f_n\}_{n\geq 1}$ converges pointwise to g on $(-1, 1)$

Let $x \in (-1, 1)$ be given.
$\lim\limits_{n\to\infty} f_n(x) = \lim\limits_{n\to\infty} x^n = 0.$
Therefore, $\{f_n\}_{n\geq 1}$ converges pointwise to g on $(-1, 1)$.

Part 2: WTS: $\{f_n\}_{n\geq 1}$ converges uniformly on $[a, b]$ for $-1 < a < b < 1$.

Let $-1 < a < b < 1$ be given.
Let $\epsilon < 0$ be given. Define $N_\epsilon = \max\{|\log_{|a|}(\epsilon)| + 1,\ |\log_{|b|}(\epsilon)| + 1\} > 0$.

Let $x \in [a, b]$ and $n \geq N_\epsilon$ be given.
$|f_n(x) - f(x)| = |x^n - 0| = |x^n| = |x|^n \leq |x|^{N_\epsilon} < \epsilon$

Thus, $\{f_n\}_{n\geq 1}$ converges uniformly on $[a, b]$ for $-1 < a < b < 1$.

Part 3: WTS: $\{f_n\}_{n\geq 1}$ does not converge uniformly on $(0, 1)$.

Suppose to the contrary that $\{f_n\}_{n\geq 1}$ does converge uniformly on $(0, 1)$.

Thus, for all $\epsilon > 0$, there exists some N_ϵ such that, $n \geq N_\epsilon \implies |f_n(x) - f(x)| < \epsilon$ for all $x \in (0, 1)$.

However, as $x \to 1$, $x^n \to 1$ regardless of how big n is.

Thus, $\{f_n\}_{n\geq 1}$ does not converge uniformly on $(0, 1)$. $\square$

4. Chapter 1, Problem 2 Let Ω be a finite set, i.e., the number of elements in Ω is finite. Let $\mathcal{F} \subset \mathcal{P}(\Omega)$ be an algebra. Show that $\mathcal{F}$ is a σ-algebra.

Ans:

Proof. Since $\mathcal{F}$ is an algebra, 4 things must be true:

(a) $\Omega \in \mathcal{F}$

(b) If $A \in \mathcal{P}$, $A^c \in \mathcal{F}$

(c) If $A,\ B \in \mathcal{P}$, $A \cup B \in \mathcal{F}$

(c') If $A,\ B \in \mathcal{P}$, $A \cap B \in \mathcal{F}$

Since $\mathcal{P}(\mathcal{F})$ is finite, $\mathcal{F}$ is finite. Suppose $n \equiv |\mathcal{F}|$, and $\{B_1, B_2, ..., B_n\} = \mathcal{F}$. Since $\mathcal{F}$ is finite, B_i is finite for all $i \in \{1, 2, ..., n\}$.

Let $m \leq n$ be given.
Suppose $\{B_1, ..., B_m\} \subset \mathcal{F}$.
Define,

- $G_1 \equiv B_1$
- $G_2 \equiv B_1 \cup B_2 = G_1 \cup B_2$ ($G_2 \in \mathcal{F}$ since $\mathcal{F}$ is an algebra)
- $G_3 = B_1 \cup B_2 \cup B_3 = G_2 \cup B_3$ ($G_2 \in \mathcal{F}$ since $\mathcal{F}$ is an algebra and $B_3, G_2 \in \mathcal{F}$)
- ...

- and $G_m \equiv B_1 \cup B_2 \cup ... \cup B_{m-1} \cup B_m = G_{m-1} \sup B_m$ ($G_m \in \mathcal{F}$ since $\mathcal{F}$ is an algebra and $B_m, G_{m-1} \in \mathcal{F}$).

It is clear to see that $G_1 \subset G_2 \subset G_3 \subset ... \subset G_m$. Thus, $\cup_{i=1}^{m} G_i = G_m \in \mathcal{F}$.

Since, $\mathcal{F}$ is an algebra, and $G_n \in \mathcal{F}$, $G_n \subset G_{n+1} \, \forall \, n \implies \cup_{i=1}^{m} G_i = G_m \in \mathcal{F}$, by *Proposition 1.1.1*, $\mathcal{F}$ is a σ-algebra. □

5. Chapter 1, Problem 4

   Let $\{\mathcal{F}_\theta : \theta \in \Theta\}$ be a family of σ-algebras on Ω. Show that $\mathcal{G} \equiv \cap_{\theta\in\Theta}\mathcal{F}_\theta$ is also a σ-algebra.

Ans:

Proof. (a) $\Omega \in \mathcal{F}_\theta$ for all $\theta \in \Theta$, since they are all σ-algebras. Thus, $\Omega \in \mathcal{G}$.

(b) Let $A \in \mathcal{G}$ be given. Thus, $A \in \mathcal{F}_\theta$ for all $\theta \in \Theta$. Therefore, $A^c \in \mathcal{G}$ since $A^c \in \mathcal{F}_\theta$ for all $\theta \in \Theta$ since they are all σ-algebras.

(c) Suppose $A_1,\ A_2 \in \mathcal{G}$. Thus, $A_1,\ A_2 \in \mathcal{F}_\theta$ for all $\theta \in \Theta$. Since $\mathcal{F}_\theta$ are all σ-algebras, $A_1 \cup A_2 \in \mathcal{F}_\theta$ for all $\theta \in \Theta$. Therefore, $A_1 \cup A_2 \in \mathcal{G}$.

    Thus, $\mathcal{G}$ is an algebra on Ω, by definition.

(d) Let $A_1, A_2, ..., A_n \in \mathcal{G}$ be given. Thus, $A_1, A_2, ..., A_n \in \mathcal{F}_\theta$ for all $\theta \in \Theta$. Since $\mathcal{F}_\theta$ are all σ-algebras, $\cup_{i=1}^{n} A_i \in \mathcal{F}_\theta$ for all $\theta \in \Theta$. Thus, $\cup_{i=1}^{n} A_i \in \mathcal{G}$.

    Therefore, $\mathcal{G}$ is a σ-algebra.

□

6. Chapter 1, Problem 6 Let Ω be a nonempty set and let $A \equiv \{A_i : i \in \mathbb{N}\}$ be a partition of Ω, i.e., $A_i \cap A_j = \phi$ for all $i = j$ and $\cup_{n\geq 1} A_n = \Omega$. Let $\mathcal{F} = \{\cup_{i\in J} A_i : J \subset \mathbb{N}\}$ where, for $J = \phi$, $\cup_{i\in J} A_i \equiv \phi$. Show that $\mathcal{F}$ is a σ-algebra.

Ans:

Proof. **Part 1:** WTS: (a) $\Omega \in \mathcal{F}$
Since $\cup_{n\geq 1} A_n = \Omega$, $\Omega \in \mathcal{F}$.

Part 2: WTS:(b) $B \in \mathcal{F} \implies B^c \in \mathcal{F}$
Let $B \in \mathcal{F}$ be given. Thus, $B = \cup_{i\in J} A_i$ for some $J \subset \mathbb{N}$.

$B^c = (\cup_{i\in J} A_i)^c = \cap_{i\in J} A_i^c$.
Since, $A \equiv \{A_i : i \in \mathbb{N}\}$ be a partition of Ω, $A_i^c = \cup_{j\neq i} A_j$.
Thus, $B^c = \cap_{i\in J} \cup_{j\neq i} A_j = \cup_{k\in \mathbb{N}/J} \in \mathcal{F}$.

Part 3: WTS: (d) $B_1, B_2, ..., B_j \in \mathcal{F}$ for some $j \in \mathbb{N} \implies \cup_{k\leq j} B_k \in \mathcal{F}$
$\cup_{k\leq j} B_k = \cup_{k\leq j}(\cup_{i\in J_j} A_i) = \cup_{m\in \cup_{n\leq j} J_n} A_m \in \mathcal{F}$.

Since (a), (b), and (d) are true, $\mathcal{F}$ is a σ - algebra. $\square$

7. Chapter 1, Problem A
Let Ω be an uncountable set (e.g. $\mathbb{R}$) and $\mathbb{C}$ be the collection of all the singleton sets (i.e. $\mathbb{C} = \{\{\omega\} : \omega \in \Omega\}$). Find the smallest σ-algebra generated by $\mathbb{C}$. Justify your answer.

Ans: $<\mathbb{C}>= \mathcal{F}_6 = \{A \subset \Omega : A \text{ is countable or } A^c \text{ is countable}\}$.
$\mathcal{F}_6$ is a σ-algebra by Example 1.1.4 in the textbook. It is the smallest σ-algebra generated by $\mathbb{C}$ because:

- While, $\emptyset, \Omega \notin \mathbb{C}$, they are in $\mathcal{F}$.
- All elements of $\mathbb{C}$ are clearly countable sets, so $\mathbb{C} \subset \mathcal{F}_6$, and thus $\{\{\omega\}^c : \omega \in \Omega\}$
- $\{\bigcup_{i\in J} A_i : \{\{\omega_i\}\}_{i\in J} \subset \mathbb{C} \text{ for some } J \subset \mathbb{R}\}$ is the set of all countable sets on Ω. Thus, it is clear to see that $\{\bigcup_{i\in J} A_i : \{\{\omega_i\}\}_{i\in J} \subset \mathbb{C} \text{ for some } J \subset \mathbb{R}\} \cup \{\bigcup_{i\in J} A_i : \{\{\omega_i\}\}_{i\in J} \subset \mathbb{C} \text{ for some } J \subset \mathbb{R}\} \cup \{\Omega, \emptyset\} = \mathcal{F}_6 \implies <\mathbb{C}>= \mathcal{F}_6$.